Hansen's paper

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<u>Setup</u>. G conn'd reductive $/E$ local field $/\mathbb{Q}_p$. Fix $\underbrace{\overline{\mathbb{Q}}_\ell \simeq \mathbb{C}}_{\text{coeff.}}$ $(\ell \neq p)$

$$\mathcal{H}(G) := C_c^\infty(G(E)) \supset \mathcal{H}(G)_0 := \{ \mathrm{Orb}_\gamma(f) = 0, \ \forall \gamma \in G(E)_{srs} \}$$

$$\overset{\text{K's density thm}}{=\!=} \{ \Theta_\pi(f) = 0, \ \forall \pi \text{ temp} \}$$

$$\begin{array}{lcl}
\mathrm{Exc}(W_E, \hat{G}) & & \\
\quad \cup\!\shortmid & & \\
\mathfrak{Z}^{\mathrm{Spec}}(G) = \mathcal{O}(Z^1(W_E, \hat{G}))^{\hat{G}} & \xrightarrow[FS]{\Psi_G} & \mathfrak{Z}(G) \\
\quad \shortparallel & \searrow \ \mathfrak{Z}^{FS}(G) & \subset \ \text{Bernstein centre} \\
\text{functions on} & \shortparallel & \\
\{\text{ss } L\text{-par}\} & \mathrm{im}(\Psi_G) & \\
 & \shortparallel & \\
\text{Spectral Bernstein centre.} & \langle \text{exc. operators} \rangle &
\end{array}$$

Only need slogans:

$\boxed{\text{FS1}}$ exc. ops commute with Hecke operators T_μ

$\boxed{\text{FS2}}$ FS construction is compatible with parabolic induction.

$$\begin{tikzcd}
\mathfrak{Z}(G) \overset{\text{last week}}{\curvearrowright} \underset{(cusp)}{\mathcal{H}(G)} \arrow[d, "\text{mod } \mathcal{H}(G)_0"] & \supset & \overset{\text{"unstable" } SO_{srs}=0}{\mathcal{H}(G)_{00}} \arrow[d] & \supset & \overset{O_{srs}=0}{\mathcal{H}(G)_0} \\
\mathfrak{Z}(G) \overset{①}{\curvearrowright} \underset{(cusp)}{I(G)} \arrow[d, "\text{mod } I(G)_{00}"] & \supset & I(G)_{00} & & \\
\mathfrak{Z}^{FS}(G) \overset{②}{\curvearrowright} \underset{(cusp)}{S(G)} & & & &
\end{tikzcd}$$

(with $\mathfrak{Z}(G) = \mathfrak{Z}(G)$ between the first two rows and $\mathfrak{Z}(G) \cup \mathfrak{Z}^{FS}(G)$ between the last two)

Lem 1 (warm up) $\mathfrak{Z}(G)$ respects $\mathcal{H}(G)_0$. $\Rightarrow$ induces action ①.

Pf. $f \in \mathcal{H}(G)_0$, $\mathfrak{z} \in \mathfrak{Z}(G)$. $\forall \pi$

$$\Rightarrow \Theta_\pi(f) = 0 \Rightarrow \Theta_\pi(\mathfrak{z} * f) \overset{\text{Varma}}{\underset{4.1.3}{=}} \mathfrak{z}_\pi \Theta_\pi(f) = 0 \Rightarrow \mathfrak{z} * f \in \mathcal{H}(G)_0. \quad \square$$

Main Th. (Hansen) $\mathfrak{Z}^{FS}(G)$ respects $I(G)_{00}$. $\Rightarrow$ induces action ②.

(ignore cusp for now)

Diagram

~~$\mathrm{Py}(\mathfrak{z}.\Theta)(f) = \Theta(\mathfrak{z} * f)$~~ Fact: $\mathfrak{z}.\Theta_\pi = \mathfrak{z}_\pi \Theta_\pi$, $\mathfrak{z}_\pi \in \mathbb{C}$

$$\mathfrak{Z}(G) \curvearrowright \underset{?}{D}(G) := \langle \Theta_\pi : \pi \text{ irred and } ? \rangle \subset I(G)^*$$

dense

$\Theta_\pi|_{G(E)_{ell}} \neq 0$

$? \in \{\phi, \text{ell}, \text{temp}\}$

$\cup$

$$\underset{?}{SD}(G) := \{\Theta \in D(G) : \Theta(g) = \Theta(g'), \forall g \underset{st}{\sim} g' \text{ srs}\} \subset S(G)^*$$

$$= \{\Theta \in D(G) : \Theta(f) = 0, \forall f \in I(G)_{00}\}$$

Lem 2 Main Th $\Leftrightarrow$ $\mathfrak{Z}^{FS}(G)$ preserves $SD_{temp}(G)$.

Pf. Arthur 96 + ε $\quad \mathfrak{z} \in$

($\Leftarrow$) $f \in I(G)_{00}$. WTS $\mathfrak{z} * f \in I(G)_{00}$.

By hypo, $\mathfrak{z} * \Theta \in SD_{temp}$ $\Rightarrow$ $(\mathfrak{z} * \Theta)(f) = 0$ $\xrightarrow{\text{Art 96}}$ $\mathfrak{z} * f \in I(G)_{00}$

$\qquad \qquad \| \; \Theta(\mathfrak{z} * f)$ $\square$

Cor. $\Theta = \sum_{i \in I} a_i \Theta_{\pi_i}$ is atomically stable

fin; π_i irred

atomically: minimal stable comb.

stable: const on stable conj. class

Then $\phi_{\pi_i} :=$ FS parameter of π_i are <u>all equal</u>.

<u>Pf</u>. If false, $\underset{\boxed{FS3}}{}$ $\exists\, \mathfrak{z} \in \mathfrak{z}^{spec}(G) = \{\text{funcs on SS par.}\}$ whose values at ϕ_{π_i} are <u>not</u> all equal.

$\boxed{FS3}$

Since $\underset{\uparrow \text{ Stable}}{\Theta}$, $\underbrace{\underset{\uparrow\, \mathfrak{z}^{FS}}{\underline{\Phi_G(\mathfrak{z})}}.\Theta}_{\text{stable by main thm.}}$ are both stable, but lin. indep. $\square$.

$\boxed{\text{Proof of Th}}$

<u>Prelim</u>. $\mathfrak{z}(G) \curvearrowright D_{temp}(G) = \overbrace{D_{ell}(G)}^{\Theta|_{G(F)_{ell}} \neq 0} \oplus \overbrace{D_{ind}(G)}^{\bigoplus_{M \subsetneq G} D_{ell}(M)^{W(M,G)}}$

$\cup \quad \cup \quad \cup \quad \cup$

$\mathfrak{z}^{FS}(G) \curvearrowright SD_{temp}(G) = SD_{ell}(G) \oplus \underline{SD_{ind}(G)}$

$\curvearrowright$: STP (Lem 2)

$SD_{ind}(G) = \bigoplus_{M \subsetneq G} SD_{ell}(M)^{W(M,G)}$

That is,

$\boxed{STP}$ $\mathfrak{z} \in \mathfrak{z}^{FS}(G)$, $\Theta \in SD_{temp}(G)$,

$\Rightarrow \mathfrak{z}\cdot\Theta \in SD_{temp}(G)$, i.e. $\underset{= \Theta(\mathfrak{z} * f)}{(\mathfrak{z}\cdot\Theta)(f)} = 0$, $\forall f \in I(G)_{00}$

<u>Step 0</u>. Reduction to elliptic, cusp. (by Arthur + induct. on ss. rank)

<u>Why?</u> If $\Theta \in SD_{ind}(G)$. <u>WMA</u>. $\Theta = i_M^G \Theta_M$, $\Theta_M \in SD_{ell}(M)$, $M \subsetneq G$ Levi

$$\begin{array}{ccccc} \exists\, \tilde{\mathfrak{z}} \in & \mathfrak{z}^{spec}(G) & \xrightarrow{\Phi_G} & \mathfrak{z}(G) \ni \mathfrak{z} & \\ \big\downarrow & \big\downarrow \text{can} & \curvearrowright & \big\downarrow \Gamma_M & \text{last week} \\ \tilde{\mathfrak{z}}_M \in & \mathfrak{z}^{spec}(M) & \xrightarrow{\Phi_M} & \mathfrak{z}(M) & \end{array}$$

$$\left(z \cdot i_M^G \Theta_M\right)(f) = i_M^G \Big(\underbrace{\underbrace{\underbrace{r_M(z)}_{\Phi_M(\tilde{z}_M)} \cdot \Theta_M}_{\text{stable on } M \text{ (by ind.)}}}_{\text{stable on } G} \Big)(f) = 0$$

$\therefore$ $\boxed{\text{WMA} \quad \Theta \in SD_{ell}(G)}$

<u>Next</u>. <u>WMA</u> $f \in I_{cusp}(G)_{00}$

because $f = f_{cusp} + \underbrace{f_{non\text{-}cusp}}$

killed by $SD_{ell}(G)$.

<u>Summary so far</u>. Reduced to showing:

<u>Prop</u> $\Theta \in SD_{ell}(G)$, $f \in I_{cusp}(G)_{00}$, $z \in z^{FS}(G)$

$$\Theta(z * f) = 0.$$

<u>Hecke ops</u> $\mu \in X_*(T)_+ = X^*(\hat{T})_+$, $T \subset B \subset G/\bar{E}$

Suppose $1 \in B(G, \mu)$, eg. $\mu = \mu_m = 4m\rho_G$, $m \in \mathbb{Z}_{\geq 1}$

FS construct $T_\mu \curvearrowright K_0(G) \rightsquigarrow T_\mu \curvearrowright D(G)$

<u>Key inputs</u>

$\boxed{\text{C. Fu}}$ (coarse version) $\frac{1}{\dim V_\mu} \cdot T_\mu$ is "stabilizing" as $\mu \to \infty$.

$\boxed{\text{HKW}}$ can compute T_μ on ell. part, well accept:

Prop $\Theta \in SD(G)$. Then $\frac{1}{\dim V_\mu} T_\mu(\Theta) = \Theta + \underbrace{\Theta'}_{D_{ind}(G)}$

Back to proof of Prop.

$\boxed{\text{Step 1}}$ Θ, z, f as in Prop

$$c_\mu := \frac{1}{\dim V_\mu} \cdot T_\mu \Theta (z * f) = \underbrace{\Theta (z * f)}_{\text{indep. of } \mu}, \quad \forall \mu \text{ s.t. } 1 \in B(G,\mu)$$

Pf. $c_\mu \overset{\text{HKW}}{=\!=} \Theta(z*f) + \underbrace{\underbrace{\Theta'}_{D_{ind}}\underbrace{(z*f)}_{I_{cusp}}}_{\overset{\shortparallel}{0}}$ $\square$

$\boxed{\text{Step 2}}$ Prove $c_\mu = 0$ by $\mu \to \infty$.

Pf. $c_\mu = \frac{1}{\dim V_\mu}(z \cdot T_\mu \Theta)(f) \overset{\boxed{\text{FS1}}}{=\!=} \frac{1}{\dim V_\mu}\left(T_\mu(\underbrace{z \cdot \Theta}_{\overset{\shortparallel}{\Xi}})\right)(f)$

$$\underset{D_{ell}(G)}{\Xi} \longmapsto \Xi^{st} \overset{\text{Fact}}{=\!=} \text{avg of } \Xi \text{ along each stable conj. class}$$

$$D_{ell}(G) \underset{\longrightarrow\!\!\!\!\rightarrow}{\longleftarrow} SD_{ell}(G)$$

via "ell. inner product"

on $D_{ell}(G)$

c. Fn

$\boxed{\text{Expect } \frac{1}{\dim V_\mu} T_\mu(\Xi) \xrightarrow[\mu\to\infty]{} \Xi^{st}}$ $\xRightarrow[\text{at } f]{\text{eval}}$ $c_\mu \longrightarrow \underset{I(G)_{00}}{\Xi^{st}(f)} = 0.$